A Unified Machine Learning Framework for Customer Retention in Indian E-Commerce

Churn Prediction · Customer Segmentation · Personalised Recommendations · Sentiment Analysis

BharatMart 360° · Databricks 14-Day AI Challenge · March 2026
Platform: Azure Databricks · Azure SQL · Azure Event Hub · ADLS Gen2

ABSTRACT

Indian e-commerce platforms are growing fast but losing customers just as fast. Bain & Company's How India Shops Online 2025 confirms India is now the world's second-largest e-retail market with over 270 million active buyers, heading toward $200 billion in sales by 2026. Yet churn rates exceed 30% per year, 75% of buyers purchase only once, and replacing a lost customer costs 5 to 25 times more than keeping one. Personalised product recommendations drive 31% of e-commerce revenue, but most platforms still show every customer the same bestseller list. Despite years of individual research on churn prediction, customer segmentation, and product recommendations, no published work connects all three in a single working system. Springer's Electronic Commerce Research (December 2024) explicitly identifies this combination as an unresolved gap in the literature. This paper presents BharatMart 360°, an end-to-end pipeline on Databricks that fills that gap. Using 72 months of synthetic Indian e-commerce data designed to mirror real market patterns, five models were built and deployed:

(1) An XGBoost churn prediction model achieving AUC = 0.6500 against a published benchmark of 0.6505 (Matuszelanski & Kopczewska, JTAECR 2022), with plain-language explanations of why each customer is at risk.

(2) A K-Means RFM segmentation model grouping 92,107 customers into Champions, Loyal Customers, and At-Risk segments, validated by three independent methods at Silhouette = 0.4108 (Wong, Tong & Haw, JTDE 2024).

(3) An ALS collaborative filtering engine generating personalised product recommendations for 97.1% of customers, performing 9.5 times better than random (Padhy et al., MDPI 2024).

(4) An RFMQ re-ranking layer using entropy-derived weights (Chen et al., Frontiers in Big Data 2025) that improves recommendation ranking quality (NDCG@10) by 25.5% over raw ALS.

(5) A Mosaic AI sentiment analysis model scoring 369,230 product reviews with zero parse failures (Roumeliotis et al., Elsevier NLP Journal 2024), identifying 40,970 hidden dissatisfied customers who gave high star ratings but wrote negative text.

The five models together identify 233 Urgent-priority Champions and, within that group, 120 who are actively complaining with specific product or delivery issues. This is a signal no individual model can produce. The pipeline runs nightly on Databricks, writing results to a live AIBI dashboard by morning. Keywords: customer churn prediction, RFM segmentation, collaborative filtering, RFMQ re-ranking, LLM sentiment analysis, Indian e-commerce, Databricks, real-time ML deployment, entropy weight method.

1. Introduction

India's e-commerce sector has crossed a turning point. Bain & Company's *How India Shops Online 2025* confirms India has overtaken the United States as the world's second-largest online retail market by number of active shoppers. Over 270 million Indians now buy products online, and the market is heading toward $200 billion in total sales by 2026. That is a massive opportunity. But this growth hides a serious structural problem underneath: platforms are bringing in new customers at enormous speed and losing them almost as fast.

Three numbers tell the full story of how bad the retention problem really is.

First, customer churn on India's biggest platforms - Amazon India, Flipkart, and Myntra is above 30% per year, according to Statista data cited in R Discovery (June 2025). That means roughly one in three customers stops buying within twelve months. These are not fringe buyers. These are customers who created an account, made a purchase, received the product, and then vanished from the platform entirely. No follow-up notification, no targeted promotion, and no personalised recommendation reached them in time.

Second, a ResearchGate study published in February 2025 by Naik et al. found that 75% of e-commerce buyers are one-time purchasers. Three out of four people who buy something never come back. They are not necessarily dissatisfied. Many of them simply forget, find a better deal elsewhere, or never had a reason to return. The platform had one chance to turn them into repeat customers and missed it. No system on any of these platforms caught them before they left.

Third, the financial cost of this problem is enormous and well-documented. Harvard Business Review, working with Bain & Company, established that getting a new customer to replace someone who left costs 5 to 25 times more than keeping the existing one. And if a platform can improve its retention rate by just 5%, profits go up by 25% to 95%. These are not theoretical numbers from a classroom exercise. They have been validated across dozens of industries over more than two decades.

Personalisation makes the problem worse when it is missing. McKinsey & Company published a study in 2024 reporting that 71% of consumers expect personalised product experiences when they shop online, and 76% say they are frustrated when they do not get them. Twilio Segment's 2024 study puts hard numbers on the revenue impact: personalised product recommendations are responsible for 31% of all e-commerce revenue. That means a platform showing every customer the same generic bestseller list, regardless of what they actually like, is leaving nearly a third of potential revenue on the table. And it is pushing customers toward competitors who personalise better.

So, the problem has four parts. First, figuring out which customers are about to leave before they actually go. Second, understanding how valuable each customer is, because not all customers are equal and not all deserve the same level of attention. Third, showing them the right products to make them stay, because a generic discount code is not a retention strategy. And fourth, understanding why they are unhappy in the first place, because a customer who is leaving because of late deliveries needs a completely different response than a customer who is leaving because of poor product quality.

Each of these four problems has been studied individually for years. Churn prediction alone has an enormous body of research. A 2025 systematic review published by MDPI and Tandfonline looked at 837 peer-reviewed churn prediction papers. RFM segmentation, which groups customers by how recently they bought, how often, and how much they spent, is a well-established technique that has been used in marketing for decades. Collaborative filtering, the technology behind every "customers who bought this also bought" recommendation, is deployed on every major e-commerce platform in the world. And sentiment analysis of product reviews has been studied extensively using both traditional methods and modern large language models.

But here is the gap that nobody has filled: nobody has connected all of these in one working system. These problems are always studied in isolation, in separate papers, by separate research teams, producing separate outputs that never talk to each other. The churn prediction team does not know about the segmentation results. The recommendation engine does not know which customers are at risk. The sentiment analysis results sit in a separate dashboard that nobody connects back to the retention pipeline. Springer's Electronic Commerce Research published a comprehensive review in December 2024 that explicitly states combining churn prediction, customer segmentation, and personalised recommendations in a single deployed framework 'fills a gap in the literature.' An independent study published on arXiv in October 2025 reached the same conclusion: this combination has not been described in prior work.

This paper presents BharatMart 360°, a pipeline built on Databricks that fills that gap. The contribution is not a new algorithm. It is the demonstration that connecting five independently validated research models on shared infrastructure produces outputs that no single model can generate alone and that those outputs run as a live system, scoring every customer every night and putting actionable results on a dashboard every morning. The most concrete example of what these produces: 120 customers who are simultaneously the most valuable segment (Champions), flagged as likely to churn, and have written negative reviews with specific complaints about delivery delays or product quality. No individual model can produce this list. All five working together do.

2. Literature Review

2.1 The Retention and Personalisation Problem in Indian E-Commerce

The size of India's customer retention challenge is confirmed by multiple independent sources. Bain & Company reports that India reached $60 billion in online sales in 2024, but year-over-year growth has slowed from above 20% to 10–12%. Part of this slowdown comes from declining repeat purchase rates. Platforms are spending more and more money on advertising and promotions to acquire new customers, but an increasing fraction of those customers buy once and never return. The money spent to acquire them is wasted.

Statista data cited in R Discovery (June 2025) puts hard numbers on the churn problem: annual churn on the three largest Indian platforms is above 30%. That is consistent with the ResearchGate finding that 75% of buyers are one-time purchasers. If you combine these two numbers, the picture is clear: Indian e-commerce platforms have a massive funnel at the top (270 million buyers) and a massive leak at the bottom (30%+ annual churn, 75% one-time buyers).

The financial consequence is established beyond debate. Harvard Business Review and Bain & Company have documented for over two decades that keeping an existing customer is 5 to 25 times cheaper than finding a new one. A 5% improvement in customer retention produces a 25% to 95% improvement in profitability. These numbers have been validated in telecommunications, banking, retail, SaaS, and e-commerce. They apply directly to Indian e-commerce, where customer acquisition costs are rising fast because Amazon, Flipkart, Myntra, and Meesho are all bidding for the same pool of digital shoppers.

McKinsey & Company published a 2024 report showing that 63% of consumers who receive generic, non-personalised shopping experiences actively consider switching to a competitor. It is not enough to send a customer a discount code. If the discount is for a product they do not care about, they ignore it. Twilio Segment (2024) measured that personalised recommendations account for 31% of total e-commerce revenue. Solving churn without solving personalisation is like patching one hole while water pours through another. A customer retained by a generic promotion but not re-engaged with products they actually want will leave in the next cycle.

2.2 Churn Prediction - What Research Has Found and What It Missed

Churn prediction is one of the most studied problems in applied machine learning. A 2025 systematic review published by MDPI and Tandfonline looked at 837 peer-reviewed churn prediction papers published between 2020 and 2024. That is more than 800 research papers in just four years, all studying the same fundamental question: can we predict which customers will leave? The most consistent finding across all 837 papers is that gradient boosting methods, specifically XGBoost, achieve the highest accuracy scores on e-commerce churn datasets. XGBoost appeared as the top or near-top performer in the overwhelming majority of studies.

But the same review found a persistent structural gap that runs through most of these papers. The majority of them build an accurate model in a Jupyter notebook and stop there. They report an AUC score, show a confusion matrix, and conclude. They do not address how to deploy the model in a live system that scores customers every day. They do not address how to make the predictions understandable to non-technical business teams who need to act on them. And they do not address how to connect the churn score to downstream actions like targeted campaigns, personalised product recommendations, or customer service escalations.

The specific paper we use as our primary benchmark is by Matuszelanski & Kopczewska, published in 2022 in the Journal of Theoretical and Applied Electronic Commerce Research. This is the most thorough published comparison of churn prediction algorithms on e-commerce data. The authors tested 24 different algorithms on the Brazilian Olist dataset with 96,180 customers and found XGBoost as the top performer with an AUC (Area Under the Curve) score of 0.6505. AUC measures how well a model can distinguish between customers who will churn and those who will stay. A score of 1.0 means perfect prediction. A score of 0.5 means the model is no better than flipping a coin. 0.6505 is a realistic score for e-commerce churn, where human behaviour is inherently messy and unpredictable.

The paper documented exactly which customer features are most predictive of churn: how much a customer spent on their first order, how many items they bought, and the cost of shipping. Importantly, they also found that review scores and review text topics did not predict churn. This seems counterintuitive at first. Why would a customer who gives five-star reviews still leave? Because satisfaction and loyalty are different things. A customer can be perfectly happy with the product they received and still never come back because they found a cheaper alternative, forgot about the platform, or simply did not need anything else. Satisfaction does not equal loyalty. This finding shaped our entire feature engineering approach, and our own SHAP analysis independently confirmed the exact same pattern on Indian data.

A paper published in ScienceDirect’s Computers in Human Behavior (August 2024) identified a specific weakness in standalone churn models that none of the 837 papers in the systematic

review had addressed. The weakness is this: churn models tell you the probability that a customer will leave, but they do not account for how valuable that customer is to the business. A customer with a 60% churn probability who spends ₹69,000 per year is far more urgent to save than a customer with the same 60% probability who spends ₹500. The churn model treats them identically. This observation is exactly why our framework integrates churn prediction with RFM segmentation. The churn model answers "who is leaving?" The segmentation model answers "how much does it matter?"

2.3 Customer Segmentation Using RFM Analysis

RFM stands for Recency (how recently did the customer buy?), Frequency (how often do they buy?), and Monetary (how much do they spend?). It is one of the simplest and most effective ways to understand customer value, and it has been used in marketing for decades before machine learning was even a term. The beauty of RFM is that business teams immediately understand it. When you tell a marketing manager "This customer is a Champion who spent ₹69,000 and has not bought in 90 days," they know exactly what to do without needing to understand any math, any algorithm, or any model architecture.

The paper we follow for our segmentation model is by Wong, Tong & Haw, published in 2024 in the Journal of Telecommunications and the Digital Economy. They applied K-Means clustering on RFM features for an online retail dataset from the UK with about 3,700 customers. They tested both K-Means and Hierarchical (Agglomerative) clustering and found that Hierarchical clustering produced slightly better-separated groups (Silhouette Score of 0.47 compared to K-Means). They used three different validation methods to determine the right number of groups - the Elbow method, Silhouette Score, and Calinski-Harabasz Index and all three agreed that three groups were the best choice. We followed the exact same three-method validation approach.

We applied this methodology on BharatMart data, which has 25 times more customers (92,107 versus 3,700). An interesting finding emerged: on our data, K-Means actually outperformed Hierarchical clustering (Silhouette 0.4127 vs 0.3866). This is the opposite of what the paper found. The reason is dataset scale. Hierarchical clustering builds a distance matrix between every pair of customers. At 92,107 customers, that matrix would require 92,107 times 92,107 times 8 bytes, which is approximately 64 GB of memory. Our compute node has 16 GB. Beyond the memory issue, K-Means draws tighter cluster boundaries on larger datasets because it has more data points to learn from. At 25 times the paper's scale, the algorithms switch positions. We documented this as a finding, not an error (discussed in Section 7.2).

An arXiv study from October 2025 explicitly recommends RFM as the segmentation layer in a unified retention framework, specifically because of its business-level interpretability. More complex methods like PCA-based dimensionality reduction or autoencoder embeddings might

produce marginally better statistical metrics, but they produce segments that business teams cannot understand or act on. The same paper identifies combining churn probability with customer segment value as a contribution not described in prior published work.

2.4 Personalised Recommendations and RFMQ Re-Ranking

Collaborative filtering is the technology behind every “customers who bought this also bought” recommendation you see on any e-commerce platform. The core idea is simple: find customers who behave similarly to you, see what they bought that you have not bought yet, and recommend those products. The most common implementation for large systems is a technique called ALS (Alternating Least Squares), which takes a massive table of customer-product interactions and learns hidden patterns in it.

Padhy, Suman, Priyadarshini & Mallick published a study in 2024 in MDPI Engineering Proceedings comparing four collaborative filtering algorithms on a dataset of over one million product ratings. They found that SVD (Singular Value Decomposition) performed best overall, but ALS had the worst accuracy among the methods tested, with an RMSE of 1.4485. More importantly, they included an explicit warning that ALS tends to overfit, it memorises the training data instead of learning general patterns unless its hyperparameters (settings) are carefully tuned. We took this warning very seriously and designed our model training process around it, testing 18 different parameter combinations systematically.

RFMQ re-ranking was introduced in a 2025 paper published in Frontiers in Big Data by Chen et al. The authors extended the classic RFM model with a fourth dimension: Quantity, meaning the average number of items a customer buys per transaction. A customer who buys one expensive item behaves differently from a customer who buys ten cheap items, even if their total spend is the same. The paper’s key innovation is using a mathematical method called information entropy to objectively determine how important each of the four RFMQ dimensions is for a given dataset. Instead of a human manually deciding “Frequency matters twice as much as Recency,” the entropy method calculates this from the data itself. Their central result showed that RFMQ re-ranking improved recommendation quality (F1 score) from 0.1709 to 0.3093 at TOP-5, which is an 81% improvement. This is the paper behind our Model 4.

2.5 Sentiment Analysis of Product Reviews Using LLMs

Traditional sentiment analysis uses small, specialised models trained on labelled datasets. These models work well for simple cases (“Great product!” is clearly positive, “Terrible quality” is clearly negative) but struggle with nuance, sarcasm, and multi-aspect opinions. Modern approaches use Large Language Models (LLMs) which can understand complex language patterns without needing task-specific training.

Roumeliotis, Tselikas & Nasiopoulos published a study in 2024 in Elsevier's Natural Language Processing Journal comparing GPT and LLaMA models specifically on e-commerce product reviews. They demonstrated that LLMs outperform traditional approaches at extracting not just overall sentiment (positive, negative, neutral) but also specific issue categories from free-text reviews. Their approach uses a structured JSON prompt: the LLM receives a review and returns a structured object with sentiment, a confidence score, and an array of specific issues like delivery delay, wrong product, or packaging damage.

This structured approach is particularly powerful for e-commerce because it goes beyond simple classification. A customer who writes "Loved the product but delivery took three weeks, very frustrating" is positive about the product itself but deeply negative about the logistics experience. A simple positive/negative classifier would lose this distinction. The structured JSON output captures both signals separately, and downstream systems can filter by specific issue type.

The most important finding from the sentiment analysis literature for our framework is that star ratings are structurally unreliable indicators of actual satisfaction. Customers frequently give high star ratings while writing negative text, and vice versa. This phenomenon, called hidden dissatisfaction, means that any analysis based solely on star ratings will miss a significant portion of unhappy customers. This connects directly to what Matuszelanski & Kopczewska found in their churn study: review ratings do not predict churn. The text inside the review might, but only if you can analyse it at scale with a model capable of understanding language.

2.6 The Gap Nobody Has Filled

The paper that validates our entire framework design is from Springer's Electronic Commerce Research, published in December 2024. This is a comprehensive review that examined all existing work on churn prediction, customer segmentation, and product recommendations. Their conclusion was direct and unambiguous: combining all three in a single unified framework 'fills a gap in the literature.' The individual pieces exist. Hundreds of churn papers exist. Dozens of RFM papers exist. Thousands of collaborative filtering implementations exist. What does not exist is a single system that connects them together and produces a unified output.

Our framework goes one step further than what Springer described as the gap. We add sentiment analysis as the fifth model, providing the "why" layer. Not just who is leaving and how valuable they are and what products to show them, but why they are unhappy, extracted from their own words in their own reviews. The combination of all five- churn prediction, value segmentation, product recommendations, segment-aware re-ranking, and sentiment analysis has no precedent in the published literature as of March 2026.

3. Research Questions

The framework is built around three research questions. Each one is tied to a documented gap in the literature and one or more models in the pipeline.

Research Question	Models	Status
RQ1 - Can we predict which Indian e-commerce customers will stop buying, matching published accuracy benchmarks, and deploy it as a nightly system producing per-customer explanations?	Model 1 (Churn)	Complete
RQ2 - Can we group customers by value and combine that grouping with churn probability to create a single priority signal that tells the business team exactly who to save first?	Models 1 + 2 (Churn + RFM)	Complete
RQ3 - Can we generate product recommendations ranked by both purchase affinity and customer value, and can we understand why at-risk customers are unhappy, from their own words?	Models 3 + 4 + 5 (ALS + RFMQ + Sentiment)	Complete

4. Methodology

4.1 Data and Synthetic Generation

The pipeline runs on 72 months of synthetic Indian e-commerce data, spanning January 2020 to December 2025. We generated the data ourselves because no public Indian e-commerce dataset exists with the combination of features we needed: six years of continuous transaction history, customer browsing sessions with page views and duration, cart events including add, remove, and Wishlist actions, product reviews with free text, shipping and delivery records, refund records, seller commissions, inventory snapshots across warehouses, and marketing campaign responses, all for the same set of 92,107 customers and 50,000 products. Public datasets like the Brazilian Olist dataset cover transactions but not sessions. Flipkart datasets cover product catalogs but not individual customer behaviour over time. We needed everything connected to every customer.

The synthetic data was generated using CTGAN (Conditional Tabular GAN), a machine learning technique that learns the statistical patterns in real data and generates new records that look realistic without containing any real customer information. We trained CTGAN models on nine publicly available Kaggle datasets covering real transactions, products, reviews, campaigns, and supply chains. The generated data preserves the original distributions, correlations between columns, and tail behaviours like extreme outliers.

We designed the data to reflect documented Indian market patterns, verified against published reports. The year-over-year growth rate of 23% matches the CAGR reported by Bain & Company. UPI (India's mobile payment system) adoption grows from 20% to 72% between 2020 and 2025, matching Reserve Bank of India published data. Diwali falls on different dates each year (October or November) based on the actual Hindu calendar, creating the festive shopping spikes visible in real Indian e-commerce data. There is a visible demand dip during the COVID-19 lockdown in early 2020. Electronics sales spike to 40% of orders during the Diwali and Big Billion Day season versus 20% in normal months. Wedding season (November to April) shows increased spending on jewellery and home furnishing categories.

We also deliberately injected seven categories of data quality problems into the raw data, because real-world production data is never clean. These include: NULL customer IDs in 2% of orders (orders floating in the system with no customer attached), duplicate rows at 1% across all tables, mixed date formats where 15% of order dates arrive as DD-MM-YYYY instead of the standard YYYY-MM-DD and 100% of review dates use DD-MM-YYYY, zero-amount transactions appearing in streaming events, ghost foreign keys where 3% of shipments, reviews, and refunds reference order IDs that do not exist in the orders table, and late-arriving dimension records where customer records appear in the system after their first transaction. These seven categories of problems are detected and handled at the Silver layer of the pipeline, demonstrating that the system can deal with messy data, not just clean data.

4.2 Data Architecture - The Medallion Pipeline

All data flows through a three-layer architecture called the Medallion pattern, managed by Databricks Unity Catalog with LakeFlow Declarative Pipelines. Think of it as a factory with three stages: raw materials come in at the Bronze layer exactly as they are, get cleaned and quality-checked at the Silver layer, and get assembled into finished business-ready products at the Gold layer.

The Medallion pattern matters for a machine learning project because data quality directly determines model quality. If dirty data reaches the model training step, every model trained on it will learn wrong patterns and make wrong predictions. By enforcing data quality at the Silver layer, every downstream model from churn prediction to sentiment analysis starts from clean, validated, trustworthy data. Changes to quality rules happen in one place (Silver) and automatically propagate to every model and dashboard that reads from it. This is the difference between a one-time analysis and a sustainable production system.

Bronze Layer - Raw Data, Exactly as It Arrives

The Bronze layer has three separate data pipelines running in parallel, each bringing data from a different source system. This mirrors how real e-commerce platforms receive data from multiple systems simultaneously.

Path 1 - Azure SQL Database: Six master data tables (customers, products, sellers, brands, categories, and warehouses) live in Azure SQL Database. When a customer updates their address or a product price changes, Change Data Capture (CDC) records that change automatically. These changes flow into Databricks Bronze tables, ensuring the pipeline always has the most recent version of every reference record.

Path 2 - Azure Event Hub (real-time streaming): Order events, payment confirmations, browsing sessions, cart actions (add to cart, remove from cart, add to Wishlist), and campaign responses stream in through Azure Event Hub using the Kafka connector. Events arrive at roughly one per two seconds during active periods. This simulates a real e-commerce platform where customer actions happen continuously throughout the day and night.

Path 3 - ADLS Gen2 (bulk file drops): Historical operational data (shipments, reviews, refunds, commissions, inventory snapshots, and geographic reference data) arrives as CSV files in Azure Data Lake Storage. Databricks AutoLoader automatically detects new files and loads them. These represent the daily batch exports that back-office operational systems typically produce.

All 17 Bronze streaming tables store data exactly as it arrived, including all seven types of injected quality problems. Nothing is cleaned, filtered, or modified at this stage. The principle is: preserve the raw truth first, fix it in the next layer. This ensures that if a data quality rule turns out to be wrong, the original data is still available.

Silver Layer - Cleaned, Validated, Trustworthy Data

The Silver layer takes the messy Bronze data and makes it reliable. 17 materialised views apply data quality rules and transformations:

Dual-source reconciliation: Orders and payments arrive from two different sources (ADLS files and Event Hub) with different column names for the same data. ADLS calls the price “order_amount” while Event Hub calls it “subtotal.” ADLS calls shipping cost “freight_value” while Event Hub calls it “delivery_charge.” The Silver layer uses COALESCE logic to check both column names and take whichever is not null, producing consistent names regardless of which source the data came from.

Date format handling: A robust date parser tries both YYYY-MM-DD and DD-MM-YYYY formats and converts everything to a proper date type. Without this, a date like “05-03-2024” could be interpreted as either March 5th or May 3rd, depending on which format the system assumes.

Deduplication: Duplicate rows are removed by keeping only the most recent row per primary key, based on the ingestion timestamp. This ensures that if the same record arrives twice from different sources, only one copy survives.

Ghost foreign key flagging: Records that reference orders that do not exist are flagged with a Boolean column called _is_ghost_order rather than deleted. This means they remain visible for data quality dashboards while being excluded from business calculations and model training.

Gold Layer - Business-Ready Star Schema

The Gold layer assembles cleaned Silver data into a star schema optimised for dashboards and machine learning. 20 materialised views include six-dimension tables (dim_date with Indian fiscal year April–March and exact festival dates per year, dim_customer enriched with geographic region, dim_product with brand and category names plus price tier, dim_seller with performance tier, dim_warehouse, dim_geo), eleven fact tables covering all major business events, and three pre-aggregated tables. The agg_customer_360 table provides one row per customer with their complete RFM history, refund statistics, and review averages. This is the foundation table that multiple ML models read from.

4.3 Model 1 - XGBoost Churn Prediction

Reference paper: Matuszelanski & Kopczewska, JTAECR 2022

Defining What “Churn” Means in E-Commerce

Before building any model, we had to answer a fundamental question: what does “churn” mean in e-commerce? In telecommunications, churn is straightforward. The customer calls the company, cancels their contract, and leaves. There is a clear moment of departure. In e-commerce, there is no contract to cancel. A customer simply stops buying. They do not send an email. They do not fill out a form. They just disappear. One day they are browsing products, and then weeks go by with no activity at all.

We defined a churned customer as someone who meets two conditions: they have not placed an order in 90 days, and they have placed at least 2 orders in their lifetime. The 90-day threshold is the industry standard for e-commerce churn, widely used in both academic studies and industry practice. The minimum 2 orders filter removes one-time buyers from the churn calculation. A person who bought once and never returned was never loyal to begin with. They represent a different business problem (acquisition and conversion) rather than retention.

Applying this definition to BharatMart's 92,107 customers: 25,373 (27.5%) are churned and 66,734 (72.5%) are active. This creates a class imbalance, there are 2.6 active customers for every churned one. If we train a model on this data without correcting the imbalance, it would learn to always predict "active" and look 72.5% accurate on paper while completely failing at its actual job: catching churners. We applied scale_pos_weight = 2.63 to XGBoost, which tells the algorithm to penalise missed churners 2.63 times more heavily than misclassified active customers.

The 12 Features - What the Model Learns From

We built 12 features from 6 Silver tables for each of the 92,107 customers. These features were chosen based on the findings from the reference paper and expanded with additional behavioural signals available in our dataset. The features fall into four categories:

Monetary behaviour (2 features): total_spent is the total amount of money a customer has spent across all their orders, measured in Indian rupees. avg_order_value is their average spend per individual order. Together these measure how much money the customer puts into the platform. A customer whose total_spent is declining over time, or whose avg_order_value is shrinking, is showing classic pre-churn behaviour.

Dissatisfaction signals (2 features): refund_count is how many times a customer has requested a refund. refund_rate is refunds as a percentage of total orders. A customer who refunds 3 out of 5 orders (a 60% refund rate) is expressing deep unhappiness with the platform, even if they never bother to leave a bad review. Refund behaviour is a stronger signal of dissatisfaction than review ratings because it costs the customer effort and time to initiate a refund.

Engagement behaviour (4 features): abandon_rate measures what fraction of cart adds never turned into purchases — the customer browsed, liked something enough to add it to their cart, but then did not buy it. avg_pages_viewed is how many pages a customer browses per session on average. avg_session_duration is how long they spend on the platform per visit, measured in seconds. session_count is their total number of visits across all time. Together these four features paint a detailed picture of how engaged the customer is with the platform.

Preference indicators (4 features): pct_organic is the share of sessions that came from organic Google search. pct_social is the share from social media platforms. avg_rating is the average star rating they give in their product reviews. payment_idx is their most frequently used payment method, encoded as a number: UPI = 0, credit card = 1, debit card = 2, net banking = 3, Cash on Delivery = 4.

Three Data Quality Fixes Discovered During Analysis

Three significant data problems were discovered and fixed during exploratory analysis, before any model was trained. These are not edge cases. Each one would have fundamentally broken the model if left unfixed.

Fix 1 - total_amount was 99% empty: The total_amount column in the orders table had valid values for only about 15,000 out of 1.6 million rows. The remaining 99% were null. This column was supposed to represent the final price the customer paid after discounts. We recomputed it as order_amount minus discount_amount, which gave every single order a valid monetary total. Without this fix, 99% of customers would have appeared to have spent zero rupees lifetime, and total_spent, the most important feature in the model would have been meaningless.

Fix 2 - Cart abandonment rate was calculated incorrectly, twice: The first attempt at calculating abandon_rate looked for a "purchase" action in the cart_events table. The result: every single one of the 92,107 customers showed an abandon_rate of 1.0, meaning 100% cart abandonment. That made no sense. The problem: there is no "purchase" action recorded in the cart_events table. Purchases are recorded in the completely separate orders table. After fixing this by using actual order count as the purchase signal, some customers showed negative abandon_rates. This happened because certain customers placed more orders than they had cart_add events — they used direct buy buttons, repeat order features, or mobile quick checkout that bypasses the cart entirely. The rate was clipped to a valid range of 0 to 1.

Fix 3 - Payment data join was returning incomplete results: The initial approach joined payment records directly to customers using customer_id. This returned data for only 13,666 out of 92,107 customers less than 15% coverage. The reason: in BharatMart's data model, payment records do not link directly to customers. They link to orders via order_id, and orders link to customers via customer_id. Joining payments to orders first, then aggregating by customer, gave all 92,107 customers their correct preferred payment method.

The Data Leakage Discovery - Why the Model's First Score Was Wrong

This was the single most important finding during the entire model development process. The first version of the model included 14 features instead of 12 and returned an AUC of 0.9922. That number looked spectacular near-perfect churn prediction. It was completely wrong.

An AUC of 0.99 means the model is almost perfectly predicting who will churn. But real human behaviour is never that predictable. If it were, every e-commerce platform would have already solved churn. A realistic e-commerce churn model should score somewhere between 0.60 and 0.70 at best, based on published benchmarks. When a model scores 0.99, the explanation is

almost always the same: the model is not learning genuine patterns in customer behaviour. It is cheating by reading the answer from its inputs.

The cheat was data leakage. The churn label is defined as: recency_days > 90 AND total_orders >= 2. Both recency_days and total_orders were included as input features. The model simply learned to check whether recency_days was above 90, which is literally the definition of churn. It was not predicting anything. It was reading the label back from the features.

The fix was to remove both recency_days and total_orders from the feature set, leaving 12 purely behavioural features that do not contain the label definition. After removal, AUC dropped from 0.9922 to 0.6500. That dramatic drop is the proof that the remaining 12 features contain real, genuine predictive signal not just a mathematical restatement of the churn label.

In production, the model needs to detect early warning signals before a customer reaches day 90. At day 55, a customer is not churned by definition, but their session duration might be dropping and their cart abandonment might be rising. The model must catch these behavioural signals while there is still time to intervene. Including recency_days as a feature would make the model wait until day 90 and then confirm the customer has already left. That is a post-mortem report, not a prediction.

Model Training, Selection, and Explainability

Three models were trained on a 70/30 stratified split (64,474 training rows, 27,633 test rows), preserving the 27.5% churn ratio in both sets:

Logistic Regression served as the baseline at AUC = 0.6412. This is a simple linear model that draws a straight boundary between churned and active customers. It cannot capture complex interactions between features, but it establishes a minimum performance bar. Notably, our Logistic Regression outperformed the reference paper's LR (which scored 0.5862) because we engineered richer features from six different data tables rather than the paper's simpler Olist feature set.

Random Forest scored AUC = 0.6307, slightly below Logistic Regression. This was an untuned model with default parameters. With proper hyperparameter optimisation it could potentially improve, but XGBoost already met the benchmark target so additional tuning of Random Forest was unnecessary.

XGBoost scored AUC = 0.6500, the best of all three. XGBoost builds decision trees sequentially, where each new tree specifically focuses on correcting the mistakes of the previous one. It captures non-linear interactions between features that linear models miss. For example, it can learn that COD payment combined with declining session_count predicts churn more strongly than either feature alone.

SHAP (SHapley Additive exPlanations) was used to make the model's decisions understandable. For every customer, SHAP calculates exactly how much each feature pushed the prediction toward churn or away from it. The top churn signals: total_spent (low spenders churn more), avg_order_value (shrinking orders), avg_pages_viewed (customers who stop browsing deeply), session_count (fewer visits), abandon_rate (cart hesitation), and payment_idx (COD users churn more). avg_rating ranked last, confirming the reference paper's finding on a different dataset. The scoring output includes a plain-text top_churn_reason field per customer so that a campaign manager can immediately see: "This customer's primary risk factor is declining session engagement and high cart abandonment."

Production Scoring - How the Model Runs Every Night

The trained XGBoost model is registered in MLflow Model Registry under the name bharatmart.ml.bharatmart_churn_model with a Production alias. Every night, the batch scoring notebook loads this model by alias, reads the latest feature table from bharatmart.ml.churn_features, scores every customer, and writes the results to bharatmart.ml.gld_churn_predictions. The output table includes customer_id, churn_probability (a number between 0 and 1), churn_risk_band (Critical, High, Medium, or Low), top_churn_reason (a plain-text explanation from SHAP), and scored_date. The entire scoring run takes under two minutes on Serverless compute for 92,107 customers.

Library version pinning is critical for production stability. The model was trained with xgboost version 2.0.3 and scikit-learn version 1.3.2. If the scoring environment has a different version installed, the model will fail to deserialise a common and frustrating production failure mode. The scoring notebook explicitly installs the pinned versions at the start of every run. This ensures the model works identically whether it is run today, next week, or next month, regardless of what other libraries are installed on the cluster.

4.4 Model 2 — K-Means RFM Segmentation

Reference paper: Wong, Tong & Haw, JTDE 2024

Model 1 tells us who is at risk of leaving. Model 2 tells us how valuable each customer is. Not all churners deserve the same level of attention: a customer who spends ₹69,000 per year is worth roughly 122 times more than one who spends ₹565. The retention team should not spend equal effort on both.

The three RFM features (Recency, Frequency, and Monetary) were normalised using Yeo-Johnson Power Transformation because the raw values are on completely different scales (2 to 2,258 days for Recency versus ₹30 to ₹827,799 for Monetary). Without normalisation, K-Means would cluster almost entirely on Monetary and ignore the other two dimensions.

Three independent validation methods all pointed to K=3: the Elbow method showed the steepest improvement at K=3, the Silhouette Score peaked at 0.4108 for K=3 (well above every other K tested), and the Calinski-Harabasz Index peaked at 71,482.9 for K=3. K-Means outperformed Hierarchical Clustering on a 10,000-row stratified sample (Silhouette 0.4127 vs 0.3866), contradicting the reference paper but consistent with the 25-times-larger dataset scale.

Segment	Count	% Base	Avg Recency	Avg Spend (₹)	Avg Orders
Champions	13,948	15.1%	2 days	₹69,207	16
Loyal Customers	46,439	50.4%	52 days	₹2,209	23
At Risk	31,720	34.4%	66 days	₹565	7

The retention_priority field crosses churn probability from Model 1 with RFM segment from Model 2. This produces 233 Urgent customers: Champions with churn probability above 0.5. Each Urgent customer's average lifetime spends of ₹69,207 is 122 times the At-Risk average of ₹565. The churn model alone cannot identify these. The segmentation model alone cannot identify these. Only the combination produces this signal, which is exactly the capability that ScienceDirect's Computers in Human Behavior (2024) identified as missing from standalone churn models.

4.5 Model 3 - ALS Collaborative Filtering

Reference paper: Padhy et al., MDPI Engineering Proceedings 2024

Models 1 and 2 answer who is at risk and how much they are worth. Model 3 answers: what specific products should each customer see to make them stay? Without personalised recommendations, every customer sees the same generic bestseller list regardless of their individual preferences, which is exactly the experience that McKinsey's 2024 study found pushes 63% of consumers to consider switching platforms.

The interaction matrix was built from three signal sources at different confidence levels: 1,574,411 clean purchase records at confidence 1.0, cart intent signals (cart_add and wishlist_add, with cart_remove excluded as a negative signal) at confidence 0.5, and high-confidence reviews (ratings 4–5 only) at confidence 2.0. The final matrix: 1,600,485 unique pairs across 91,634 customers and 47,030 products at 99.96% sparsity.

A 6-month data window was originally planned but rejected after analysis showed that 85.1% of customers had fewer than 5 interactions within 6 months (average 2.9 per customer). ALS cannot

learn from 2–3 data points. Full 72-month history gave 17.5 interactions per customer on average with only 8.6% below 5. The confidence weight system handles recency naturally.

CrossValidator tested 18 parameter combinations across 3 folds (54 ALS fits, approximately 2.5 hours), directly responding to the reference paper's overfitting warning. Best parameters: rank=10, regParam=0.01, maxIter=10, all the minimum values tested. On a near-binary 99.96% sparse matrix, simpler models generalise better. Cold-start customers (fewer than 3 orders, 2.9% of the base) receive popularity-based fallback recommendations with a flag distinguishing them from personalised results.

Two technical adaptations were needed for Serverless production: PySpark's recommendForAllUsers() is blocked on Serverless clusters, so recommendations are computed via direct numpy matrix multiplication (U @ V.T). An initial bug using argpartition for top-10 selection was caught and fixed argpartition does not guarantee ordering within the selected items. After switching to argsort, inversion count across all 912,210 output rows was zero.

4.6 Model 4 - RFMQ Entropy-Weighted Re-Ranking

Reference paper: Chen et al. (2025). Frontiers in Big Data

Model 4 is a re-ranking layer that adjusts ALS recommendation scores based on each customer's RFMQ profile. ALS tells you which products a customer is likely to buy. RFMQ adjusts how aggressively to promote those products based on the customer's overall value and current engagement level.

The fourth dimension, Quantity (average items per cart session), was nearly identical across all three segments: Champions 3.30, Loyal 3.30, At Risk 3.29. On BharatMart, basket size does not separate customer types. The entropy weight method computed from the data confirmed this: Q received near-zero weight (0.049) while Monetary dominated at 0.774. The paper's dataset showed the opposite pattern (Q = 0.386, M = 0.145). This is a genuine cross-dataset finding Indian general e-commerce is a Monetary-first market, while the paper's outdoor sports dataset was a Frequency + Quantity market.

Dimension	Paper Weight	BharatMart Weight	What This Tells Us
R (Recency)	0.079	0.003	Most customers bought recently - R barely separates them
F (Frequency)	0.390	0.174	Moderate variation in order frequency
M (Monetary)	0.145	0.774	Spend ranges ₹29 to ₹827,799 - biggest differentiator
Q (Quantity)	0.386	0.049	Basket sizes nearly identical across all segments

Two deviations from the paper were documented. First, RFMQ is applied as a per-customer multiplier on ALS output rather than replacing the CF input matrix, because BharatMart's ALS already uses a richer three-tier confidence system. Second, segment-scoped CF was not implemented because splitting the already 99.96% sparse matrix into three smaller subsets would degrade results at this scale.

4.7 Model 5 - LLM-Powered Sentiment Analysis

Reference paper: Roumeliotis, Tselikas & Nasiopoulos, Elsevier NLP Journal 2024

Models 1 through 4 answer who is at risk, how valuable they are, and what products to show them. Model 5 answers the final question: why are they unhappy? This is the causal layer that connects departure risk to specific operational issues that the business can actually fix.

Star ratings are not enough to answer this question. Our analysis confirmed what the churn reference paper found: review ratings do not predict churn. More importantly, 11.1% of reviews showed hidden dissatisfaction, customers who gave 4 or 5 stars but wrote negative text. These customers look happy in the ratings column. They are not happy. Any system that relies solely on star ratings would miss them entirely.

We used Databricks Mosaic AI's ai_query() to run Meta's LLaMA 3.3 70B Instruct model on every product review that contained text. The model received a structured prompt asking it to return a JSON object with three fields: sentiment (positive, neutral, or negative), a confidence score between 0.0 and 1.0, and an array of specific issue categories extracted from the text. The prompt was tested on 5 sample reviews first, verified for accuracy and valid JSON structure, then run on the full 369,230-review dataset.

The structured JSON approach is critical for production use. A simple positive/negative classifier gives one label per review. The structured output gives three pieces of information: what the customer feels overall (sentiment), how confident the model is (score), and exactly what specific problems they mentioned (issues array). A review that says “Great product but terrible delivery experience, arrived two weeks late with damaged packaging” produces: sentiment = negative, score = 0.88, issues = [delivery_delay, packaging_damage]. The issues array is what makes this actionable, the logistics team can filter for delivery_delay issues specifically and see every customer who complained about it, regardless of their overall sentiment or star rating.

The choice of LLaMA 3.3 70B Instruct was driven by availability on Databricks Mosaic AI and the model’s demonstrated ability to follow structured output instructions consistently. The ai_query() function handles all infrastructure concerns, the model is served by Databricks’ managed endpoint, the notebook only needs to call the function with the prompt and receive the response. No model hosting, no GPU management, no scaling decisions.

411,470 total reviews existed in the Silver table. 380,635 (92.5%) had actual text, much higher than the 40% estimate in the design document. After removing 30,835 rating-only reviews (customers who clicked a star but wrote nothing) and 12,286 ghost-order reviews, 369,230 were scored. Processing took approximately 40 minutes on Serverless Free Edition, limited by ai_query() rate throttling. In a paid workspace with higher throughput limits this would be significantly faster. Each scored review was then joined to churn predictions and RFM segments, creating a single table where every review row contains the customer’s sentiment, churn risk, segment, retention priority, and specific complaint issues.

4.8 Production Pipeline Architecture

All model scoring runs through a scheduled LakeFlow Jobs DAG (Directed Acyclic Graph) at 2AM IST daily. The dependency chain: Bronze ingestion (CDC + Streaming + AutoLoader in parallel) → Silver cleaning → Three parallel branches: (1) ALS scoring, (2) Churn features → Churn scoring → RFM segmentation → RFMQ re-ranking, (3) Gold refresh. Sentiment scoring runs independently off Silver reviews.

Training notebooks run on ML Personal Compute once per month and are not in the daily pipeline. Scoring notebooks run on Serverless compute nightly. Every model is loaded by @Production alias from Unity Catalog never by hardcoded version numbers. Library version

pinning (xgboost==2.0.3, scikit-learn==1.3.2) prevents serialisation failures. Scalers are loaded from MLflow artifacts using transform() not fit_transform() to keep normalisation consistent.

5. Results

5.1 Model 1 - Churn Prediction Results (RQ1)

Benchmark: AUC = 0.6505 - Matuszelanski & Kopczewska, JTAECR 2022

Model	Our AUC	Paper AUC	Notes
Logistic Regression	0.6412	0.5862	Our LR beats paper's - richer features
Random Forest	0.6307	-	Untuned
XGBoost ★	0.6500	0.6505	Matched within 0.0005

Risk Band	Threshold	Customers	% of Base	Action
Critical	≥ 0.75	1,139	1.2%	Immediate intervention required
High	0.50 – 0.75	58,422	63.4%	Targeted outreach within 7 days
Medium	0.25 – 0.50	17,998	19.5%	Monitor and flag for next campaign
Low	< 0.25	14,548	15.8%	Stable - no action needed

An AUC of 0.6500 against the published benchmark of 0.6505 is effectively identical. The reference paper tested 24 algorithms on Brazilian data. We tested 3 on Indian data. The same algorithm won, with the same accuracy, on a completely different market. Both studies find monetary behaviour as the strongest churn predictor and review ratings as the weakest. This cross-dataset reproducibility is the strongest form of methodological validation.

5.2 Model 2 - RFM Segmentation Results (RQ2)

Reference: Wong, Tong & Haw, JTDE 2024

Method	K=2	K=3	K=4	K=5	Winner
Elbow (Inertia drop)	65,742	Steepest drop	Lower	Lower	K=3
Silhouette Score	0.338	0.4108	0.347	0.335	K=3
Calinski-Harabasz	58,241	71,482.9	64,103	60,887	K=3

Algorithm	Silhouette	Calinski-Harabasz	Selected?
K-Means (K=3)	0.4127	7,869.0	YES - production model
Hierarchical / Ward (K=3)	0.3866	6,344.1	No
Paper result	Hierarchical wins	-	Opposite at 25× scale

Priority	Definition	Count	% Base	Significance
Urgent	Champions + churn ≥ 0.5	233	0.3%	Each = ~122 At Risk by LTV
High	All High/Critical risk	47,946	52.1%	Structured campaign needed
Medium	Medium churn, Loyal/At Risk	17,541	19.0%	Watch list
Low	Low churn, any segment	26,387	28.6%	No immediate action

The 233 Urgent customers are the primary proof that the unified framework produces value its components cannot achieve individually. Each Urgent customer's average lifetime spends of ₹69,207 is 122 times the ₹565 average for At Risk customers. Acting on those 233 before the remaining 1,100+ Critical-risk customers who might only spend ₹565 is a resource allocation decision that no individual model can support.

5.3 Model 3 - ALS Collaborative Filtering Results (RQ3 Partial)

Reference: Padhy et al., MDPI 2024

Parameter	Grid Tested	Best Value
rank (hidden factors)	{10, 20, 50}	10
regParam (regularisation)	{0.01, 0.1, 1.0}	0.01
maxIter	{10, 20}	10
Test RMSE	Paper (explicit): 1.4485	Ours (implicit): 1.2003
Precision@10	-	0.0019 (9.5× random)

Coverage	Count	Share
ALS recommendations (personalised)	87,772 customers	97.1%
Popularity fallback (cold start)	3,449 customers	2.9%
Total (Top-10 per customer)	91,221 customers	100%

Precision@10 of 0.0019 means 9.5 times better than random guessing on a 47,030-product catalog. The modest absolute number is a structural property of precision metrics on large, sparse catalogs where many relevant recommendations overlap with the training period. Model 4 then applies RFMQ re-ranking over these candidates to improve the ranking order.

The output table, bharatmart.ml.gld_als_recommendations, contains 912,210 rows (10 recommendations per customer). Each row includes the customer_id, the recommended product_id, the rank within that customer's list (1 through 10), the raw ALS score, and a rec_source flag that distinguishes personalised ALS recommendations from popularity-based fallback recommendations. The rec_source flag is important for downstream analysis because popularity fallback recommendations are not personalised and should not be evaluated the same way as ALS-generated ones.

The model is registered at bharatmart.ml.bharatmart_als_model with a Production alias. The batch scoring notebook refits the StringIndexer on current data each night (handling new customers and products that arrived since training), extracts user and item factor matrices from the trained model, computes scores via numpy matrix multiplication, and writes the results. The

entire scoring process for 91,634 customers takes approximately 8 minutes on Serverless compute, the majority of which is spent on the matrix multiplication rather than data loading.

5.4 Model 4 - RFMQ Re-Ranking Results (RQ3 Continued)

Reference: Chen et al. (2025). Frontiers in Big Data

Metric	Raw ALS	RFMQ Re-Ranked	Improvement
Precision@10	0.0023	0.0023	+2.0%
NDCG@10	0.0002	0.0003	+25.5%

The +25.5% NDCG improvement is the primary result. NDCG rewards getting the right product to rank 1 instead of rank 9. Precision@10 only asks whether the product appeared somewhere in the top 10. RFMQ re-ranking specifically improves the ordering of recommendations. The +2.0% Precision lift confirms RFMQ did not harm the recommendation selection. The +25.5% NDCG lift confirms it improved the ordering the right products reach customers earlier in their list.

Urgent customers receive a mean RFMQ weight of 0.62, lower than High-priority customers at 0.78. This is correct: Urgent Champions have deteriorating Recency, so their composite score reflects current disengagement even though Monetary remains strong. The previous handcrafted approach would have given all Urgent Champions a flat 1.5× boost regardless of actual current behaviour. The entropy-derived weights are more honest and more useful for targeting.

5.5 Model 5 - Sentiment Analysis Results (RQ3 Complete)

Reference: Roumeliotis et al., Elsevier NLP Journal 2024

Metric	Count	Percentage
Reviews scored	369,230	100%
Positive sentiment	275,596	74.6%
Neutral sentiment	26,253	7.1%
Negative sentiment	67,381	18.2%
Parse failures	0	0.0%
Hidden dissatisfied (4-5 star + negative text)	40,970	11.1%
Urgent customers with negative reviews	120	-

Zero parse failures across 369,230 reviews. Every single ai_query() call returned valid JSON that the extraction logic could parse. This validates the structured prompting approach at production scale.

Hidden Dissatisfaction - 40,970 Customers Rating-Only Analysis Misses

40,970 reviews (11.1%) gave 4 or 5 stars but the text was classified as negative. These customers clicked a high star rating but wrote complaints in their review text. A customer who writes "Product was fine but the thing shown is completely different from what we get" might click 5 stars because delivery arrived on time, but the text reveals false advertising concerns and product mismatch issues. Any analysis relying on star ratings alone would classify this customer as satisfied. They are not. The star rating is misleading. The text is truthful.

This finding independently validates what both the churn prediction study (Matuszelanski & Kopczewska) and our own SHAP analysis found: satisfaction ratings are structurally unreliable in e-commerce. The ratings are unreliable, but the text is not. If you can analyse that text at scale, you can see what the ratings hide.

The 120 Urgent + Negative Customers - The Sharpest Action List

Of the 233 Urgent-priority customers from Models 1 and 2, 120 have also left a negative review with specific complaints in the issues array. That is more than half the Urgent group actively telling the platform exactly why they are unhappy.

These 120 customers are the single highest-priority group in the entire BharatMart customer base. They are the most valuable segment (Champions), the most likely to leave (churn probability above 0.5), and they are telling the platform what went wrong (through the issues array: delivery delays, product mismatches, packaging damage, quality complaints). No individual model produces this list. Model 1 alone finds 1,139 Critical-risk customers but cannot tell which are Champions or what they complained about. Model 2 alone finds 13,948 Champions but has no departure signal. Model 3 generates recommendations but does not know who is at risk. Model 5 finds 67,381 negative reviews but cannot distinguish a complaining Champion from a complaining At Risk customer. All five together produce the 120.

Negative Reviews Do Not Predict Churn by Themselves

A surprising finding from the cross-model analysis: the negative review rate is almost identical across all churn risk bands. Critical: 17.2%. High: 18.1%. Medium: 18.5%. Low: 18.3%. Customers

at high churn risk leave negative reviews at exactly the same rate as stable customers who are not going anywhere.

This confirms what the churn model's SHAP analysis and the reference churn study both found: dissatisfaction expressed in reviews does not predict departure. A customer can be angry in their review and keep buying. A customer can be perfectly happy and never return. Negative sentiment alone is not the retention signal. The combination of high churn risk plus negative sentiment is. That is what the 120 Urgent + negative customers represent a signal that requires all five models working together to produce.

6. Real-Time Deployment

The 2025 systematic review of 837 churn papers found that real-time deployment is the most consistently absent feature. Papers build models and publish results. They do not deploy them into systems that run every day and produce actionable output. The arXiv study (October 2025) describes combining real-time scoring with churn, segmentation, and recommendations as something not previously described in the literature.

BharatMart 360°'s entire pipeline runs automatically every night at 2AM IST without any human intervention. By morning, the AIBI dashboard shows executive retention KPIs, customer risk distribution by band, segment breakdown with Urgent/High/Medium/Low priority counts, per-customer drilldown with churn probability, RFMQ-ranked product recommendations, and recent review sentiment with specific complaint issues.

A Genie Space natural language interface sits on top of the same tables. A marketing manager can type "show me all Urgent-priority customers in Mumbai who complained about late delivery" and get a filtered, actionable list without writing a single line of SQL.

MLflow Model Registry manages all production models by alias. When a retrained model is promoted to Production, the next nightly run loads it automatically. Library version pinning prevents cross-version serialisation failures that commonly break ML pipelines in production. The system is designed to run indefinitely with zero manual maintenance.

The practical impact of this deployment architecture is that every Monday morning, the retention team opens a dashboard that was updated just hours ago. They do not need to run any scripts, check any notebooks, or ask any data scientists to produce a report. The pipeline ran at 2AM. By 8AM, the dashboard shows: how many Urgent customers exist this week, which of them left negative reviews, what specific issues they complained about, and what products Model 4 recommends showing them. The team can start calling customers at 9AM with complete context

including what the customer bought, how much they have spent, why the churn model thinks they might leave, and what they said in their most recent review.

This is the operational gap that the systematic review of 837 papers identified as missing. Most published models live in Jupyter notebooks and produce AUC scores. BharatMart 360° produces a Monday morning action list.

7. Discussion

7.1 On the AUC Benchmark Match

An AUC of 0.6500 against the published 0.6505 is a difference of 0.0005 on different data from a different country. The reference paper tested 24 algorithms and found XGBoost first. We tested three and found the same. Both studies find monetary behaviour as the dominant churn signal and review ratings as the weakest. This cross-market consistency is strong evidence that the feature engineering approach transfers.

7.2 On the K-Means vs Hierarchical Deviation

The reference paper found Hierarchical Clustering better on 3,700 UK customers. Our data shows the opposite on 92,107 Indian customers (Silhouette 0.4127 vs 0.3866). The explanation is dataset scale: Hierarchical clustering requires a distance matrix of 92,107 squared entries, approximately 64 GB. At 25 times larger scale, K-Means draws tighter boundaries and the algorithms switch positions. This is a documented empirical finding.

7.3 On the Entropy Weight Cross-Dataset Finding

Chen et al. found Frequency and Quantity dominant on outdoor sports data. BharatMart shows the opposite: Monetary dominates at 0.774, Quantity is near-zero. The entropy weight method is not a fixed formula to copy. It is a diagnostic tool that reveals which customer dimensions matter most on a specific platform. A sports equipment platform segments by buying frequency and basket size. An Indian general e-commerce platform segments by spending amount. Applying the wrong weights produces poor results. The entropy method correctly adapts to each dataset automatically.

7.4 On the Unified Signal From 92,107 to 120

The progression demonstrates the framework's value: 92,107 total customers → 1,139 Critical churn risk (Model 1) → 233 Urgent Champions (Model 1 + 2) → 120 Urgent Champions who are

actively complaining with specific issues (Models 1 + 2 + 5). Each model narrows the list and adds a dimension of understanding. The marketing team can call 120 customers and know exactly who they are, why they are at risk, and what they are unhappy about. Model 3 ensures those customers see personalised product recommendations. Model 4 ensures those recommendations are ranked by actual behavioural profile, not a generic algorithm.

7.5 On Hidden Dissatisfaction

The 40,970 hidden dissatisfied customers (11.1% of reviews) confirm that star ratings are structurally unreliable in e-commerce. This independently validates the churn study's finding that review quality does not predict churn. The implication for any retention system: do not use star ratings as a satisfaction proxy. Use the text. And use a model capable of extracting structured meaning from free text at scale.

7.6 On Practical Resource Allocation

A real-world retention team has a finite budget and a finite number of hours per week. They cannot call 92,107 customers. They probably cannot call 1,139 Critical-risk customers either. But they can absolutely call 120. The framework's job is to make sure those 120 are the right 120, the customers where intervention has the highest expected value. Calling a Champion who spends ₹69,000 and is about to leave has dramatically higher expected value than calling an At-Risk customer who spends ₹565 and was already disengaged.

Moreover, the issues array from Model 5 tells the team what to say when they call. If a customer complained about late delivery, the response might be a guaranteed express shipping upgrade. If they complained about product quality mismatch, the response might be a replacement or a curated recommendation of higher-rated products in the same category. Generic "we miss you" emails are not a retention strategy. Specific, issue-aware outreach is.

This is the operational translation of the academic gap that Springer's Electronic Commerce Research described. The review asked what a unified system looks like. The answer is: it looks like a team with a short list, sorted by priority, with context on every name, updated every morning.

8. Current Limitations

The synthetic data has not been formally validated against a real Indian e-commerce dataset. Key distributional parameters were designed to match published aggregates, but a statistical equivalence test has not been conducted.

AUC does not weight false negatives by customer lifetime value. Missing a Champion who is about to churn should carry higher cost than missing an At-Risk customer. A cost-sensitive evaluation framework is planned.

The RFMQ Precision@10 lift of +2.0% is modest. RFMQ optimises for rank quality (NDCG), not hit count. A/B testing with real users would provide more comprehensive evaluation but is not possible on synthetic data.

The sentiment model (LLaMA 3.3 70B) was used through ai_query() without fine-tuning on Indian e-commerce language. Domain-specific fine-tuning, particularly for Hindi-English code-mixed reviews common on Indian platforms, could improve accuracy further.

The 120 Urgent + negative customers are identified by review text alone. Customers who are unhappy but did not write a review (the silent majority) are not captured by Model 5. Incorporating additional behavioural signals like increasing returns or declining session engagement could extend the “why” layer beyond review-writing customers.

9. Conclusion

India’s e-commerce retention problem is well-documented and financially significant. This paper presented BharatMart 360°, a unified machine learning framework addressing four connected dimensions of that problem: predicting who is about to leave, measuring how valuable they are, surfacing products relevant enough to make them stay, and understanding why they are unhappy from their own words.

Five models are complete with verified results:

Model 1: XGBoost churn prediction at AUC = 0.6500 against the published benchmark of 0.6505 (Matuszelanski & Kopczewska, 2022), with SHAP providing plain-language explanations per customer.

Model 2: K-Means RFM segmentation at Silhouette = 0.4108, K=3 confirmed by three independent methods (Wong, Tong & Haw, 2024). Three segments with clear commercial separation. Combined with churn probability to produce 233 Urgent-priority Champions.

Model 3: ALS collaborative filtering with 97.1% customer coverage and Precision@10 at 9.5 times random (Padhy et al., 2024).

Model 4: RFMQ entropy-weighted re-ranking (Chen et al., 2025) improving NDCG@10 by +25.5%. Cross-dataset finding: Indian general e-commerce is a Monetary-first market.

Model 5: LLM sentiment analysis (Roumeliotis et al., 2024) scoring 369,230 reviews with zero parse failures. 40,970 hidden dissatisfied customers identified. 120 Urgent Champions with specific complaints extracted.

The progression tells the complete story: 92,107 customers → 1,139 Critical churn risk → 233 Urgent Champions → 120 Urgent Champions who are actively complaining with specific operational issues. Each model narrows the list and adds understanding. The final 120 is a Monday morning action list with names, risk levels, lifetime value, personalised product recommendations, and specific complaint details.

Springer's Electronic Commerce Research asked in December 2024 what a unified system combining churn prediction, segmentation, and recommendations looks like in production. BharatMart 360° is that answer extended with sentiment analysis as the causal layer, deployed as a nightly automated pipeline, and validated against five independent research benchmarks.

The contribution of this work is not any single model. Each model individually matches or closely approaches its reference paper's benchmark, but individually they are incremental contributions to well-studied problems. The contribution is the connection: the demonstration that combining five models on a shared data infrastructure, scoring the same customer base nightly, and writing outputs to a single dashboard produces commercially actionable signals that the academic literature has identified as missing.

For the Indian e-commerce industry specifically, BharatMart 360° shows what a data-driven retention system looks like when the pieces actually connect. India's 270 million online shoppers deserve platforms that notice when they are leaving, understand what they value, recommend products they actually want, and listen to what they say in their reviews. The technology to do all of this exists today, runs on cloud infrastructure available to any company, and can be deployed in weeks, not years. The gap was never the technology. The gap was the integration. This paper closes it.

10. References

Section A - Industry Evidence

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